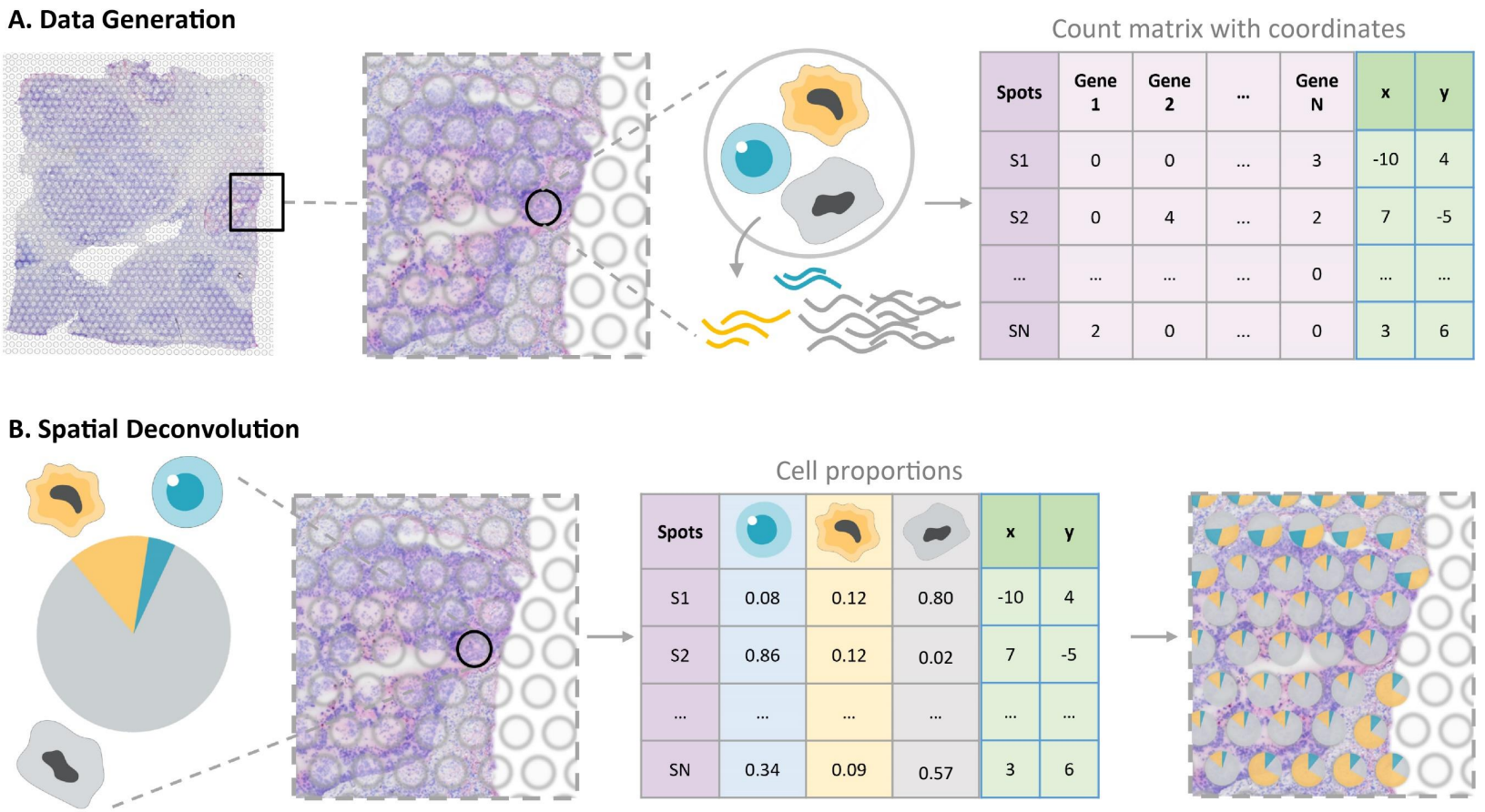
What's in a spot: the role of spatial deconvolution

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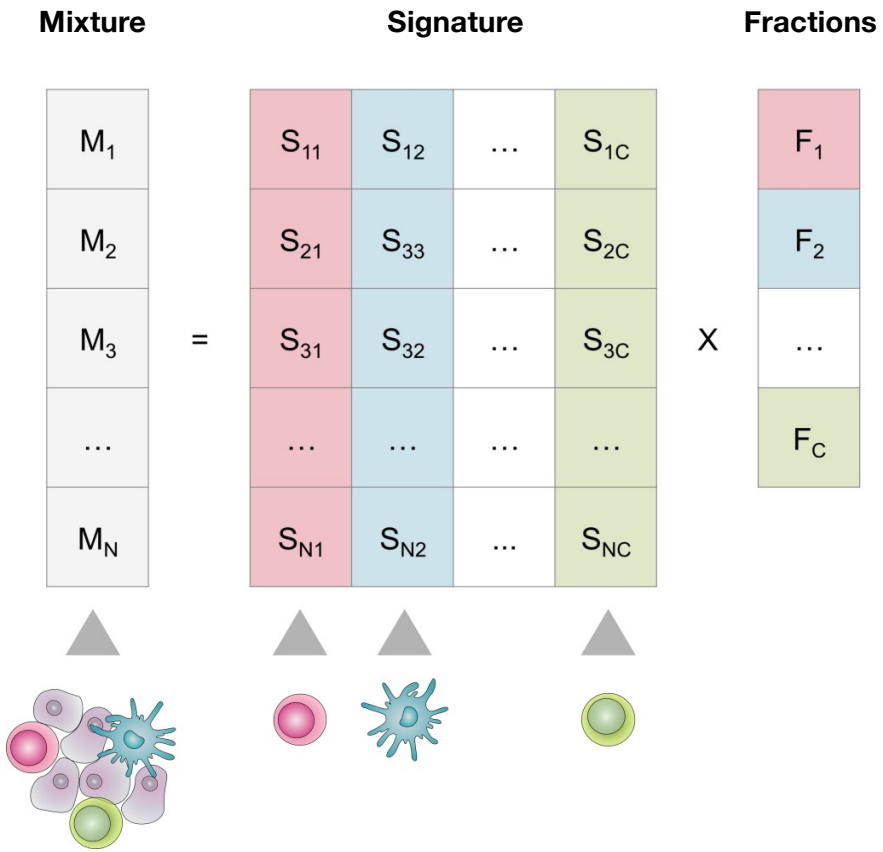
MOPITAS Autumn School | November 12-14, 2025

Deconvolution of spatial transcriptomics



Merotto L et al. Int Rev Cell Mol Biol. 2024

Deconvolution of transcriptomic data



$$\mathbf{M} = \mathbf{S} \times \mathbf{F}$$

$$M_1 = S_{11} F_1 + S_{12} F_2 + S_{12} F_3 + \ldots + S_{1C} F_C$$

$$M_2 = S_{21} F_1 + S_{22} F_2 + S_{13} F_3 + \ldots + S_{2C} F_C$$

$$\ldots$$

$$M_N = S_{N1} F_1 + S_{N2} F_2 + S_{N3} F_3 + \ldots + S_{NC} F_C$$

Deconvolution methods

- Signature matrix
- Machine/statistical learning method to solve the inverse problem

Finotello F and Trajanoski Z, Cancer Immunology, Immunotherapy, 2018

First-generation deconvolution

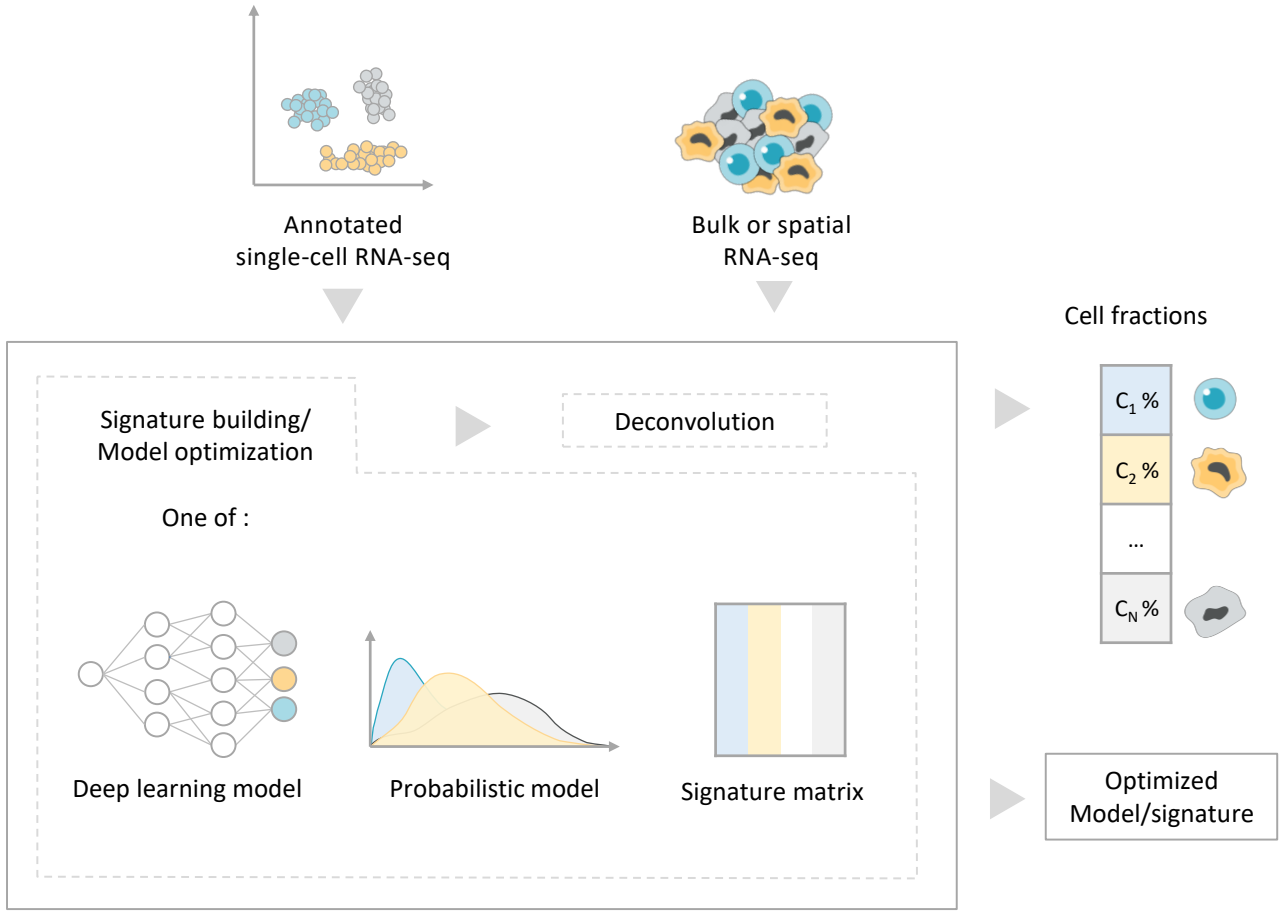


1st-gen deconvolution: pre-computed signatures (human immune cells)
immunedeconv: R interface to 1st-gen. methods for immune-cell deconvolution or scoring

Tool	Abbrev.	Type	Score	Comparisons	Algorithm	Cell types	Reference
CIBERSORT	CBS	D	Immune cell fractions, relative to total immune cell content	Intra	ν-support vector regression	22 immune cell types	Newman *et al.* (2015)
CIBERSORT abs. mode	CBA	D	Score of arbitrary units that reflects the absolute proportion of each cell type	Intra, inter	ν-support vector regression	22 immune cell types	Newman *et al.* (2015, 2018)
EPIC	EPC	D	Cell fractions, relative to all cells in sample	Intra, inter	constrained least square regression	6 immune cell types, fibroblasts, endothelial cells	Racle *et al.* (2017)
MCP-counter	MCP	M	Arbitrary units, comparable between samples	Inter	mean of marker gene expression	8 immune cell types, fibroblasts, endothelial cells	Becht *et al.* (2016)
quanTIseq	QTS	D	Cell fractions, relative to all cells in sample	Intra, inter	constrained least square regression	10 immune cell types	Finotello *et al.* (2017)
TIMER	TMR	D	Arbitrary units, comparable between samples (not different cancer types)	Inter	linear least square regression	6 immune cell types	Li *et al.* (2016)
xCell	XCL	M	Arbitrary units, comparable between samples	Inter	ssGSEA (Hänzelmann *et al.*, 2013)	64 immune and non-immune cell types	Aran *et al.* (2017)

Sturm G, Finotello F, Petitprez F, Zhang JD, Baumbach J, Fridman WH, List M, Aneichyk T. Bioinformatics, Volume 35, Issue 14, 2019, i436–i445

Second-generation deconvolution



- scRNA-seq-informed deconvolution
- Applicable to any cell type, tissue, organism, and disease
- But signatures are not validated and accuracy depends on input data and application

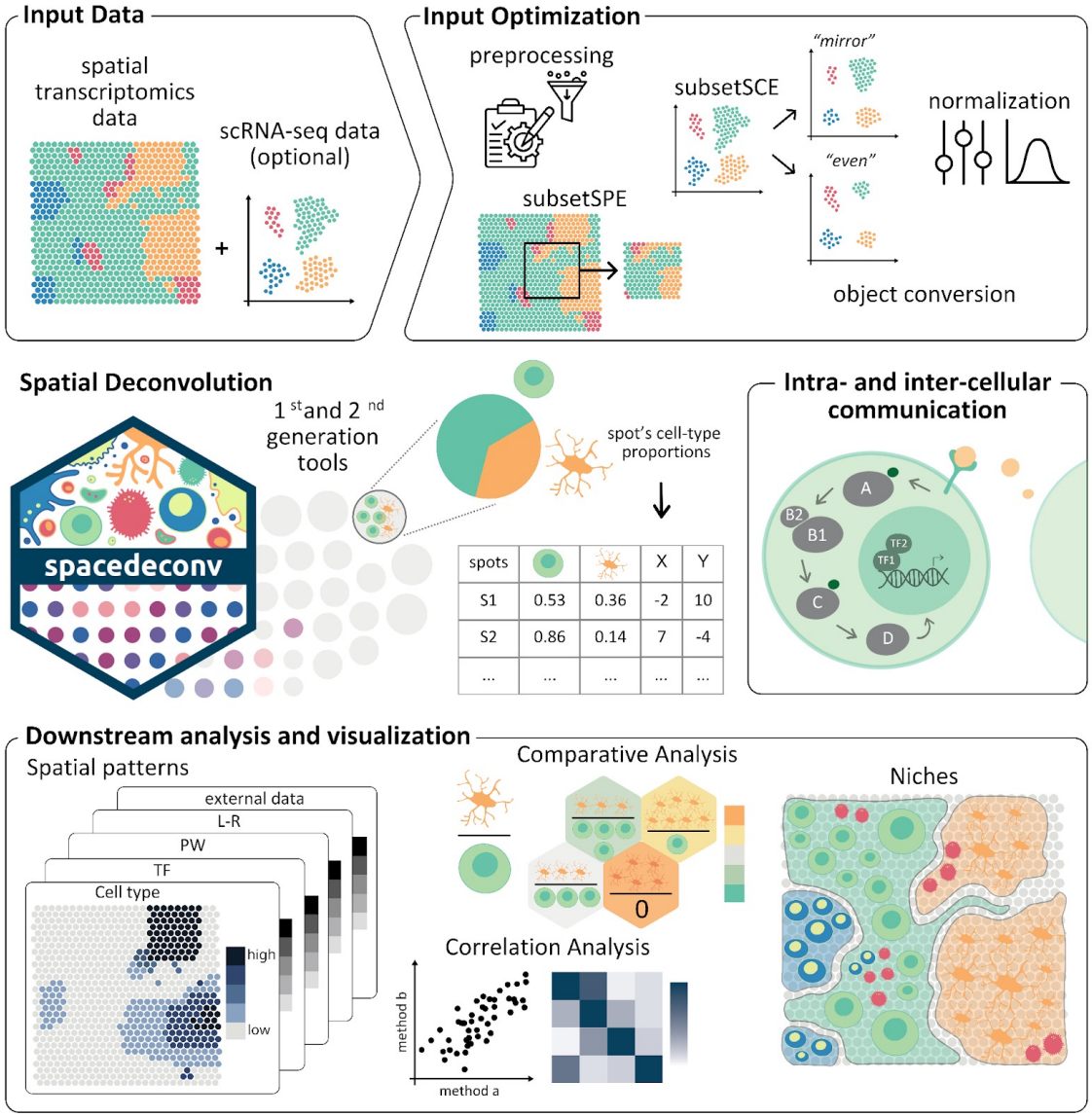
Merotto L et al. Int Rev Cell Mol Biol. 2024

Spatial deconvolution with spacedeconv

R package for unified access to spatial deconvolution methods

- 2nd-gen: cell2location, DOT, RCTD, spatialDWLS, SPOTlight, CARD
- 1st-gen: immunedeconv (13 tools)

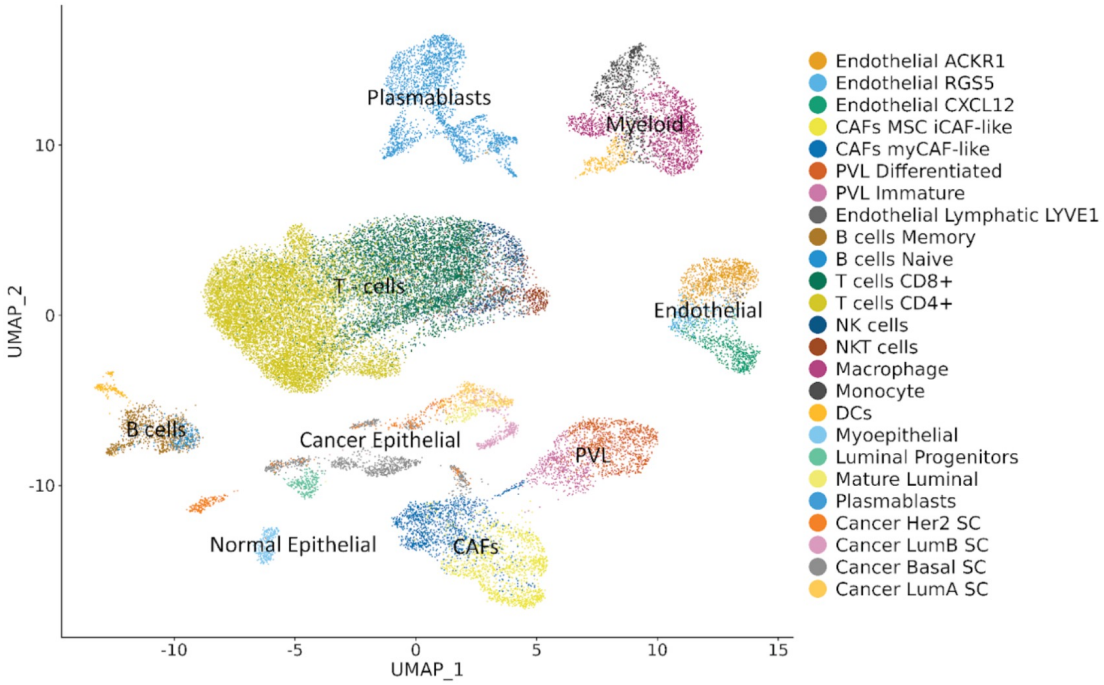
+ Modules for data preprocessing, exploratory analysis, and visualization



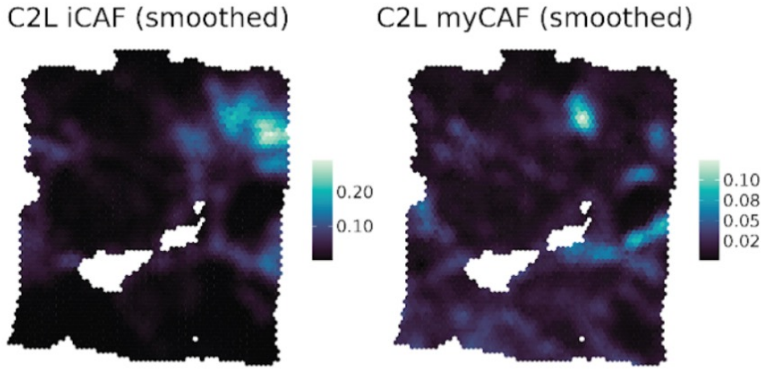
Preprint: https://www.researchsquare.com/article/rs-5102166/v1

R package: https://github.com/omnideconv/spacedeconv

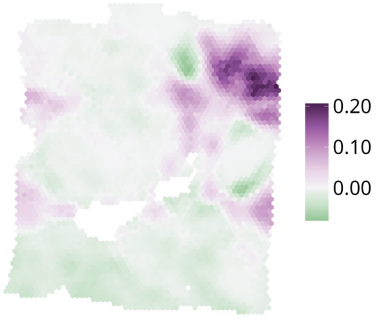
Single-cell-informed deconvolution



Training with scRNA-seq reference
→ fine-grained chart of tissue composition

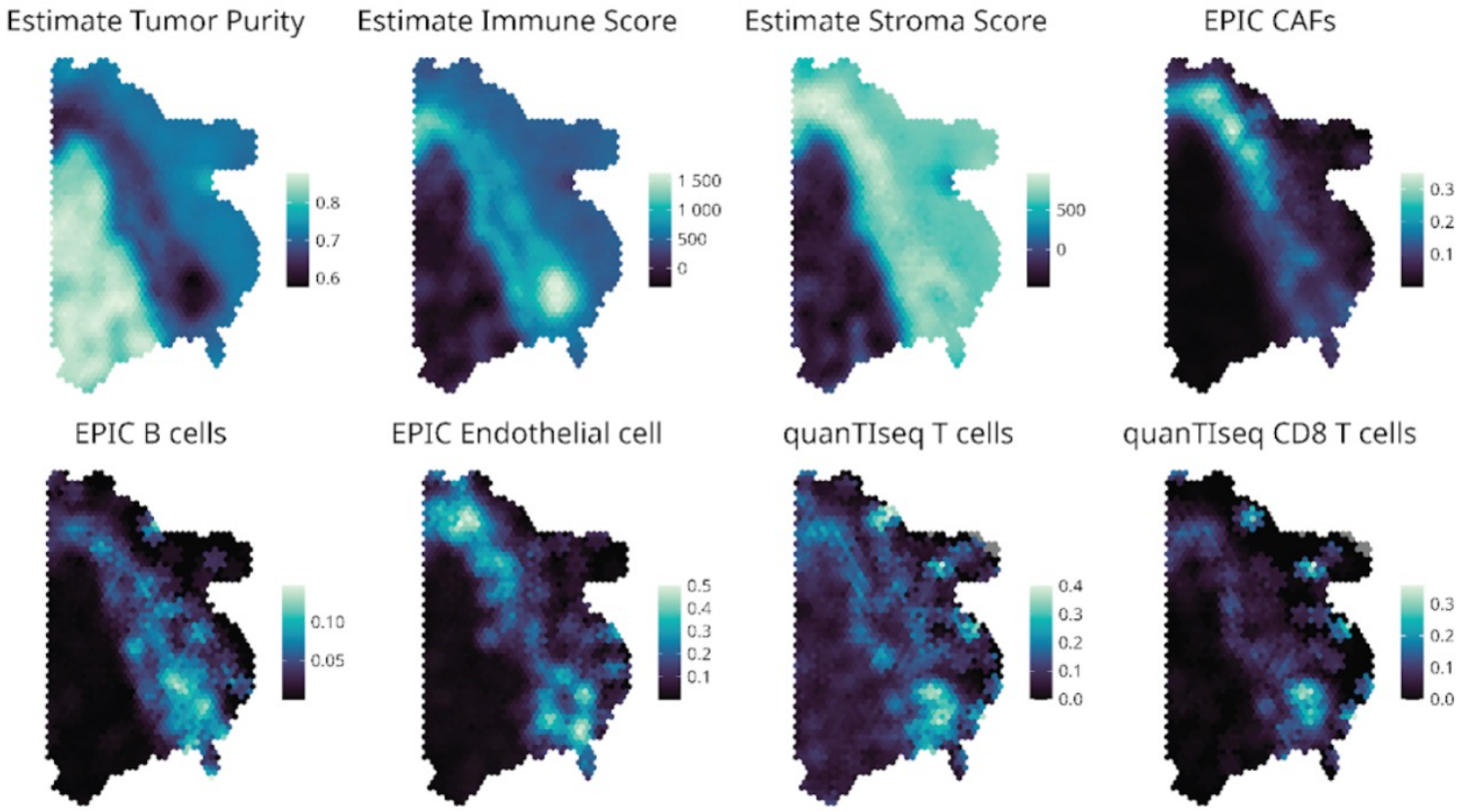


C2L iCAF vs. myCAF (smoothed)

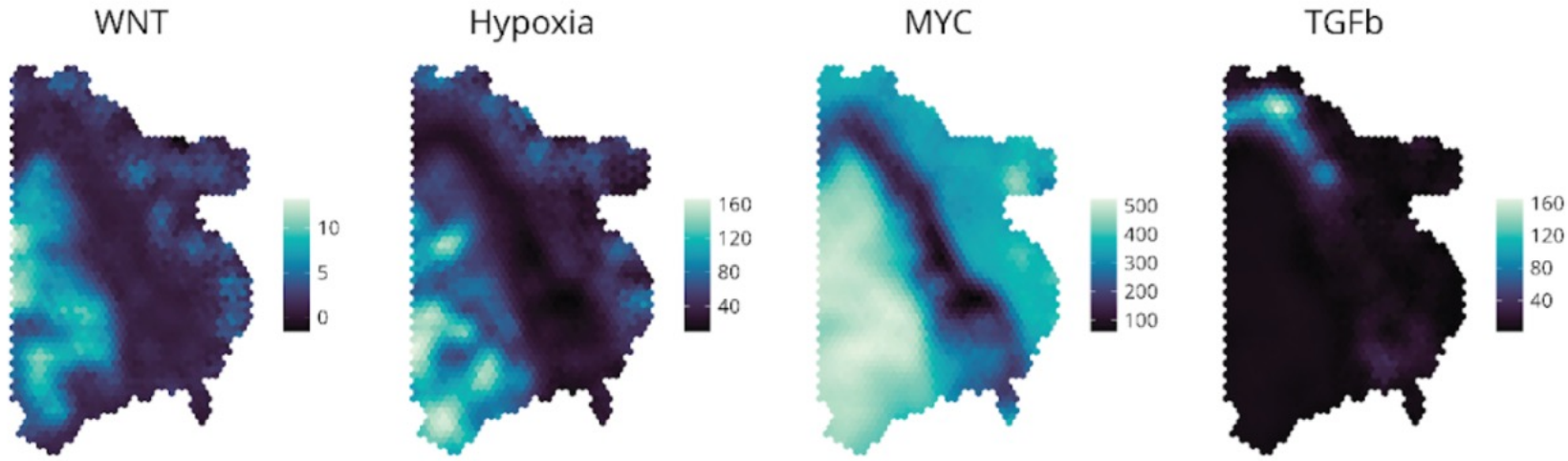


First-generation deconvolution

1st-generation deconvolution and gene-set-based methods
→ useful in case of missing/suboptimal single-cell reference

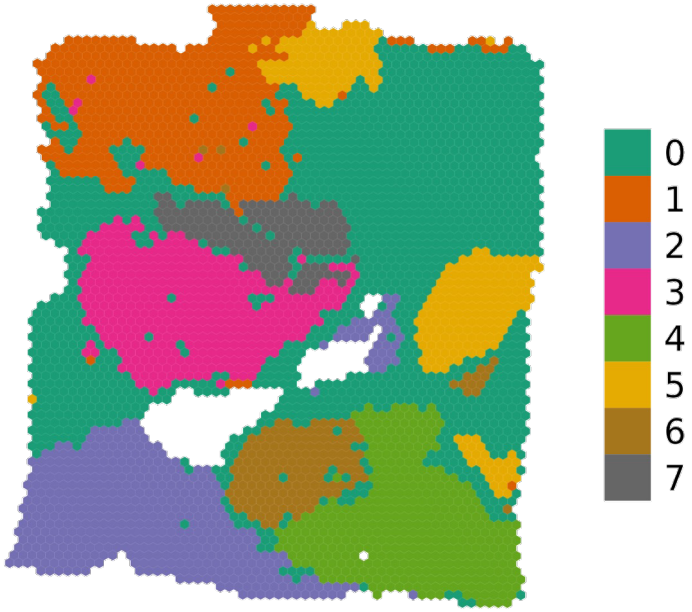


Quantification of pathway and TF activity



Quantification of pathway and transcription factor activity with decoupleR (Badia-i-Mompel et al. 2022)
→ ratifies and complements cell-type quantification

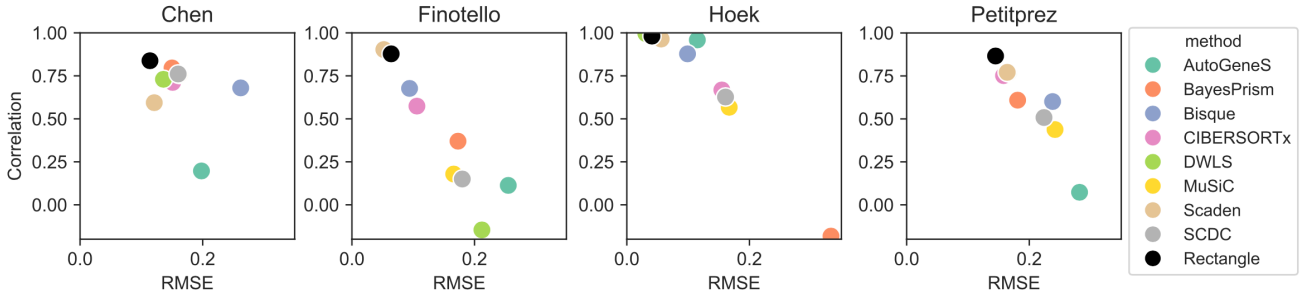
Advanced investigation of tissue niches

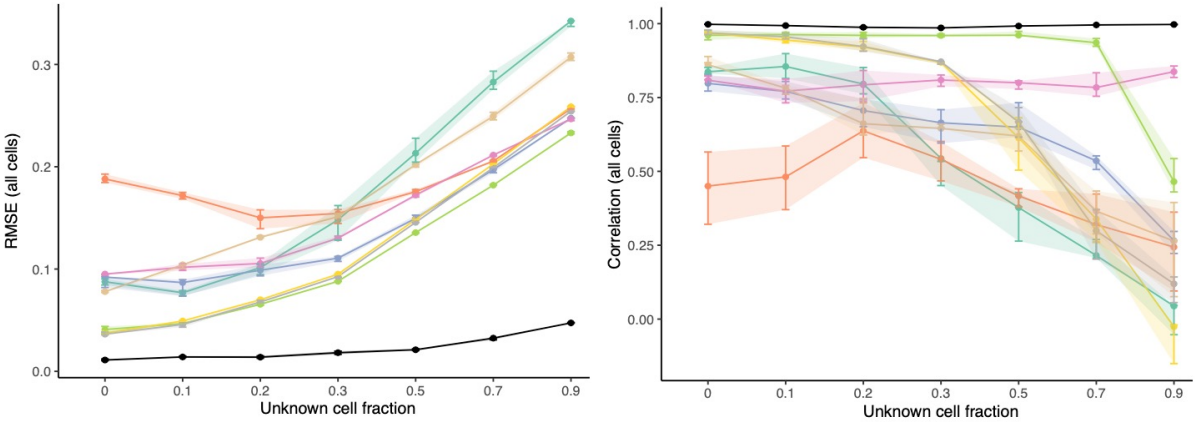


	Cell types	Pathways	TF
0	CAFs, CD4⁺ T cells, plasmablasts, NK T cells, macrophages	TGFb, TNFa, NFkB, Hypoxia, EGFR	TCF3, TOX3, SPI1, SMAD3, STAT6
3	Cancer cell, luminal cells, CD8⁺ T cells, NK cells, DCs	Trail, JAK-STAT, NFkB, TNFa, WNT	STAT2, LHX2, ZBTB33, IRF9, IRF1

- Clustering of tissue niches based on gene expression, deconvolution, or activity data
- Identification of niche-associated features

Rectangle: accurate 2nd-generation deconvolution





Rectangle

Robust algorithm for single-cell-informed deconvolution (python, scverse)

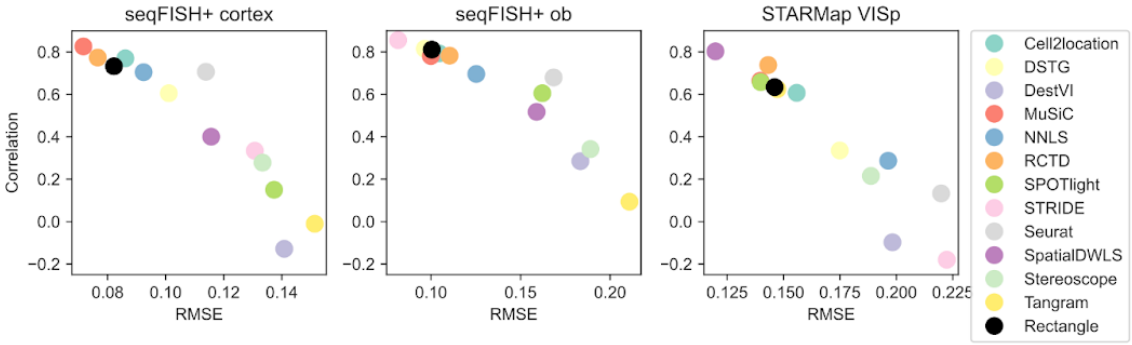
Accurate thanks to the modeling of mRNA bias, cell hierarchy, unknown content

Computationally light

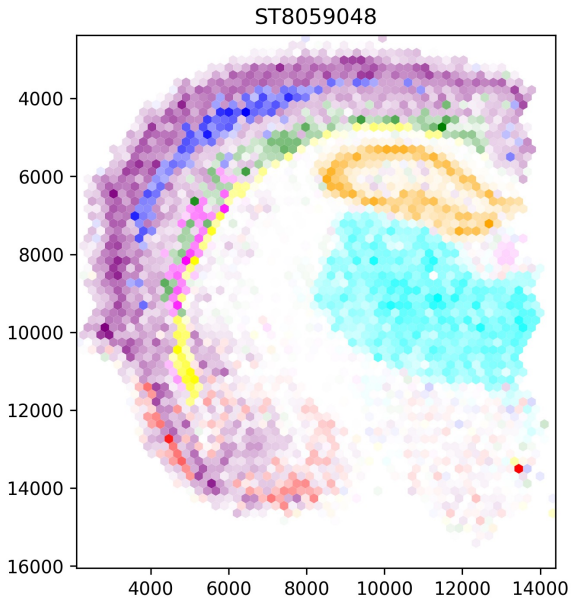
method: autogenes, bayesprism, bisque, cibersortx, dwls, music, rectangle, scaden, scdc

Eder B et al. (unpubl.), https://github.com/ComputationalBiomedicineGroup/Rectangle

Spatial deconvolution with Rectangle



Benchmarking data and framework (Spotless) from Sang-aram et al. (eLife 2024) → Rectangle among the top-performing methods across all three datasets



Fine-grained deconvolution of different excitatory neurons subtypes in the mouse brain (snRNA-seq + 10x Visium from Kleshchevnikov et al. 2022)

Conclusions

Spatial deconvolution of transcriptomics data is a powerful technique to investigate tissue architecture and its cellular and molecular underpinnings

Importance of integrating different data modalities (bulk, single-cell, spatial) and computational methodologies (e.g., 1st-gen and 2nd-gen deconvolution, activity inference) to obtain more robust insights

We can unlock the deconvolution of any cell type, tissue, organism… but we need to:

- Better understand/optimize second-generation methods
- Treasure the lessons learned from first-generation tools (e.g., robustness to unknown cellular content)

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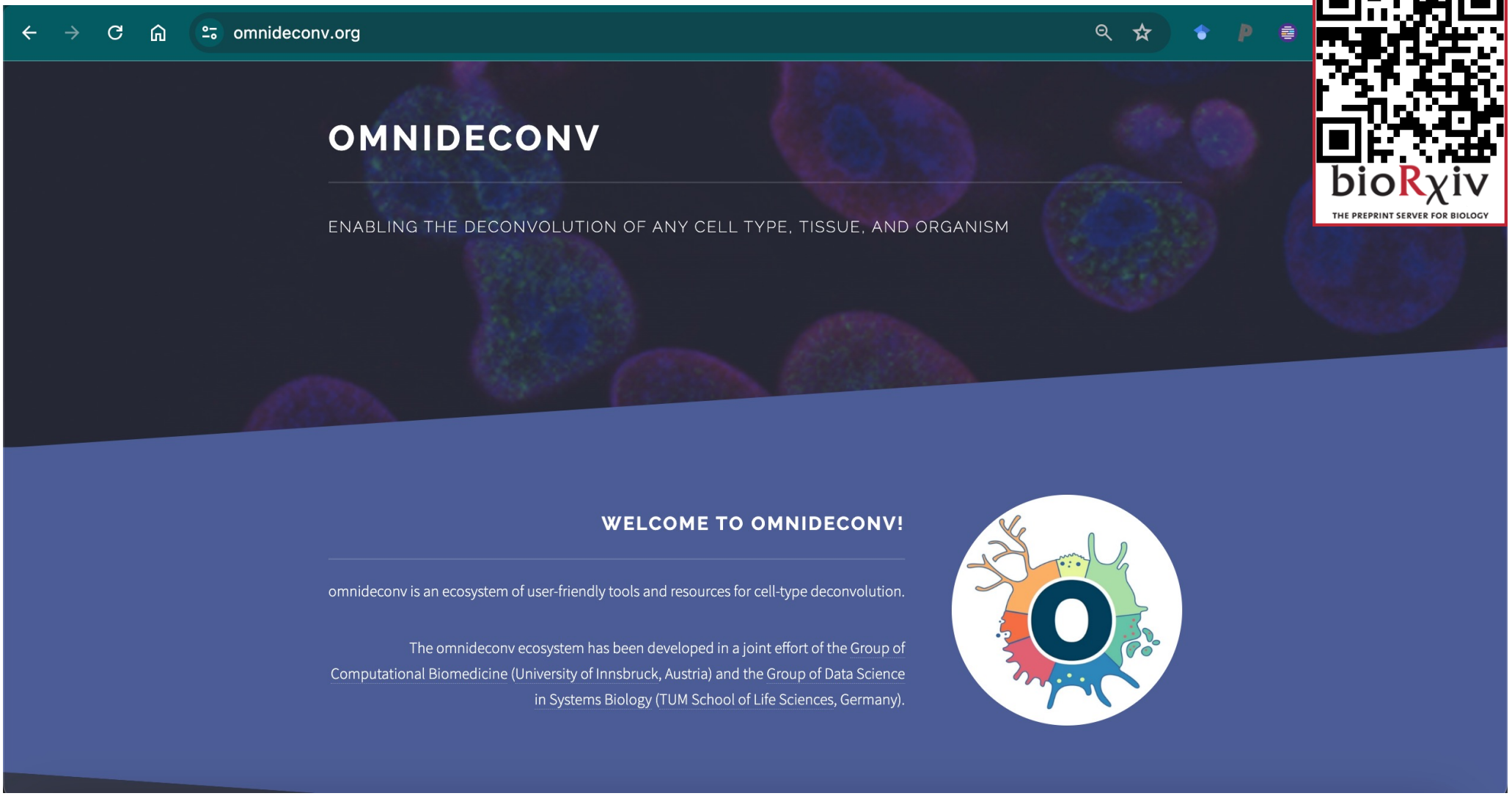
FWF Der Wissenschaftsfonds.

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